Mathieu-Hill Equations

BSA, 2024

In[396]:=

```
ClearAll["Global`*"]
```

Initial Data

In[397]:=

```
n = 5;
k = 2;
```

In[399]:=

```
Eq = x''[t] + (δ + ϵ * Cos[t]) * x[t] == 0
```

Out[399]=

```
(δ + ϵ Cos[t]) x[t] + x''[t] == 0
```

Hill Determinants

In[400]:=

```
ComputeA1[δ_, ϵ_, n_] := Table[If[i == j, δ - (i - 1)^2, If[i == 1 && j == 2, ϵ, If[Abs[i - j] == 1, ϵ / 2, 0]]], {i, n}, {j, n}];
ComputeA2[δ_, ϵ_, n_] := Table[If[i == j, δ - i^2, If[Abs[i - j] == 1, ϵ / 2, 0]], {i, n}, {j, n}];
ComputeA3[δ_, ϵ_, n_] := Table[If[i == 1 && j == 1, δ - 1/4 + ϵ / 2, If[i == j, δ - (2 * i - 1)^2 / 4, If[Abs[i - j] == 1, ϵ / 2, 0]]], {i, n}, {
ComputeA4[δ_, ϵ_, n_] := Table[If[i == 1 && j == 1, δ - 1/4 - ϵ / 2, If[i == j, δ - (2 * i - 1)^2 / 4, If[Abs[i - j] == 1, ϵ / 2, 0]]], {i, n}, {
```

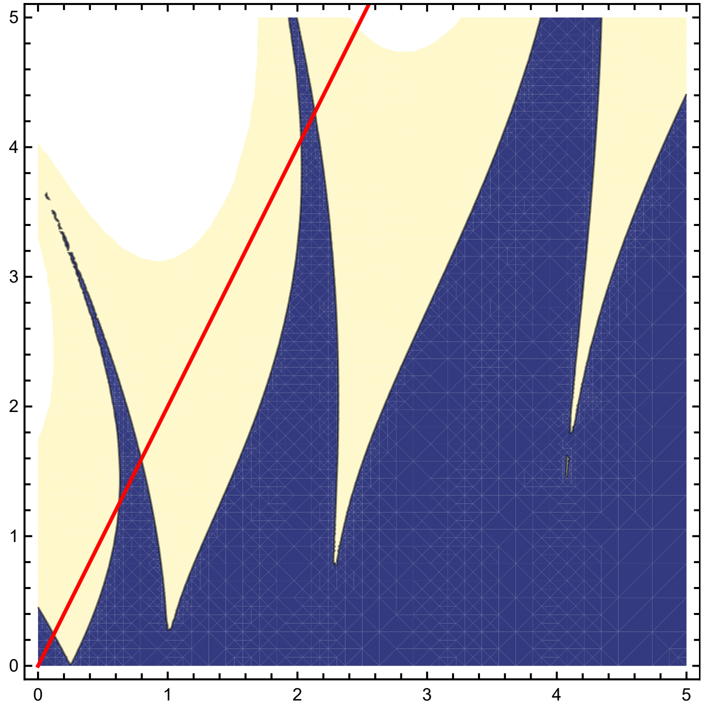
In[404]:=

```
F[δ_, ϵ_, n_] := Det[ComputeA1[δ, ϵ, n]] * Det[ComputeA2[δ, ϵ, n]] * Det[ComputeA3[δ, ϵ, n]] * Det[ComputeA4[δ, ϵ, n]];
```

In[405]:=

```
p1 = ContourPlot[F[δ, ϵ, n], {δ, 0, 5}, {ϵ, 0, 5}, Contours → {0}, PlotPoints → 20];
p2 = Plot[k * δ, {δ, 0, 5}, PlotStyle→Red];
Show[p1, p2]
```

Out[407]=



Stability Intervals

In[408]:=

```
SolDelta = Flatten[NSolve[{F[δ, k * δ, n], -0.1 < δ < 3}, δ]]
```

Out[408]=

```
{δ → 0, δ → 0.121527, δ → 0.623819, δ → 0.796839, δ → 2.02718, δ → 2.13548}
```

In[409]:=

```
StabInt = Table[{SolDelta[[i, 2]], SolDelta[[i + 1, 2]]}, {i, 1, Length[SolDelta], 2}]
```

Out[409]=

```
{{0, 0.121527}, {0.623819, 0.796839}, {2.02718, 2.13548}}
```

Plot Solution

In[410]:=

```
PlotSol[δi_, ϵi_, L_, i_] := (
    X = NDSolve[{Eq /. {δ → δi, ϵ → ϵi}, x[0] == 0, x'[0] == 0.1}, x[t], {t, 0, L}][[1]][[1, 2]];
    p1 = Plot[X, {t, 0, L}, GridLines → Automatic, AxesLabel → {t, x[t]}, PlotRange → All, PlotLabel→Style[Round[StabInt[[i, All]], 0.0001], 10],
        ImageSize→Scaled[0.8], AspectRatio → 1];
    p2 = Show[ContourPlot[F[δ, ϵ, n], {δ, 0, 5}, {ϵ, 0, 5}, Contours → {0}, PlotPoints → 20, ImageSize → Scaled[0.7]],
        Plot[k * δ, {δ, 0, 5}, PlotStyle→Red], Graphics[{Yellow, PointSize[Large], Point[{δi, ϵi}]}]];
    GraphicsRow[{p1, p2}]
);
```

In[411]:=

```
Manipulate[Manipulate[PlotSol[δ, k * δ, 80, i], {δ, StabInt⟦i, 1⟧ - 0.01, StabInt⟦i, 2⟧ + 0.01, Appearance → "Open"}], {i, {1, 2, 3}}]
```

Out[411]=